

The Compatibility Report

How the verdict is calculated

The seven checks a donor–recipient pairing passes through, the decision table that turns them into one of four verdicts, the two numbers the report does produce — and why it deliberately does not produce a single compatibility score.

Scope Direct donor-to-recipient compatibility check — `match_pipeline` and the report it generates

Companion *Paired Kidney Exchange* — *System Reference* covers the swap-matching subsystem

Audience Transplant clinicians · academic supervision · hospital IT

Reference data HLA frequencies `grifoni-colombo-2018-v1` (n=714) · LKDPI Massie 2016

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— Contents

1	There is no single score	What the report actually produces
2	Before a check can run	Five verification gates
3	The seven steps	Which ones can stop the check
4	Steps 1-3	Blood group, sensitisation, HLA mismatch
5	Steps 4-6	cPRA, donor-specific antibodies, crossmatch
6	Step 7 — the risk level	The one additive calculation
7	The decision table	Six rules, four verdicts
8	LKDPI — a separate index	Thirteen coefficients, and what they mean
9	Worked examples	Four real cases from the test suite
10	What the clinician sees	The report screen
11	What is stored, and why	Snapshotting a clinical decision
12	Reference data and provenance	Every constant, sourced
13	Known limitations and defects	Stated plainly
14	Test coverage	What is verified, and how

1 There is no single score

The most important thing to understand about this report is what it deliberately does not do. It does not reduce a donor–recipient pairing to a compatibility percentage. The verdict is **categorical**, reached by passing through a series of gates, and the reasoning behind that choice is written into the code itself:

FROM THE SOURCE

"Exactly four verdicts, deliberately no fifth and no numeric composite score: a single 0–100 number across ABO / mismatch / DSA / crossmatch / cPRA would imply a precision and a set of relative weights this pipeline has no clinical basis for."

There is no defensible way to say that one HLA mismatch is worth, say, 8% of a positive crossmatch. A composite number would have to assert exactly that, and would then be read as though it meant something. Instead each check reports its own result in its own units, and a decision table combines them by rule rather than by arithmetic.

What the report does produce

OUTPUT	TYPE	RANGE
Verdict	Categorical — the headline	Compatible · Proceed with caution · Cannot assess · Not compatible
Overall status	Categorical — where the pipeline ended	Six values: four halts, plus awaiting-crossmatch and completed
Final risk level	Categorical, from a 0–3 point sum	Low · Low-Average · High-Average · High Risk — or none
cPRA	Percentage	0–100% — how hard this patient is to match generally
LKDPI	Index, higher is worse	A donor-quality comparison. Never affects the verdict

Only one of these is arithmetic: the 0–3 point sum behind the final risk level (§6). Everything else is either a rule, a probability calculation, or a published external model.



Figure 1 — The four verdicts. Note that *cannot assess* is deliberately styled in neutral blue rather than red. It means the system lacks information, not that the pairing is bad — the interface enforces that distinction so a missing HLA row is never mistaken for a clinical finding.

2 Before a check can run

A compatibility check cannot be started on data that a clinician has not confirmed. Five flags are required, and all five must be set:

REQUIREMENT	COVERS
Patient details verified	Name, date of birth, blood type
Donor details verified	
Patient HLA typing verified	The typing that mismatch counting depends on
Donor HLA typing verified	
Patient antibody profile verified	The bead-specificity chart the virtual crossmatch reads

If any is unset the request is refused outright, and **no report row is created** — there is no partial or provisional record. The refusal names what is missing:

"Cannot run compatibility check: the patient's HLA typing, the patient's antibody/bead-specificity profile came from OCR extraction and hasn't been confirmed by a doctor yet. Review and re-save it against the source document first."

This is the seam between the extraction pipeline and the clinical one. Machine-read values may be stored and reviewed, but they cannot reach a compatibility verdict until a human has signed them off.

A separate read-only readiness endpoint lets the interface preview what is missing before a doctor commits to running anything. In that preview an unverified antibody profile is treated as non-blocking — the wizard can still be started, and verification happens on the way through.

3 The seven steps

The pipeline runs in a fixed order. Four steps are **gates** — failing one abandons every later step, and the report records exactly where it stopped. Two steps are **informational**: they compute values that appear in the report but cannot stop it. The last classifies what survived.

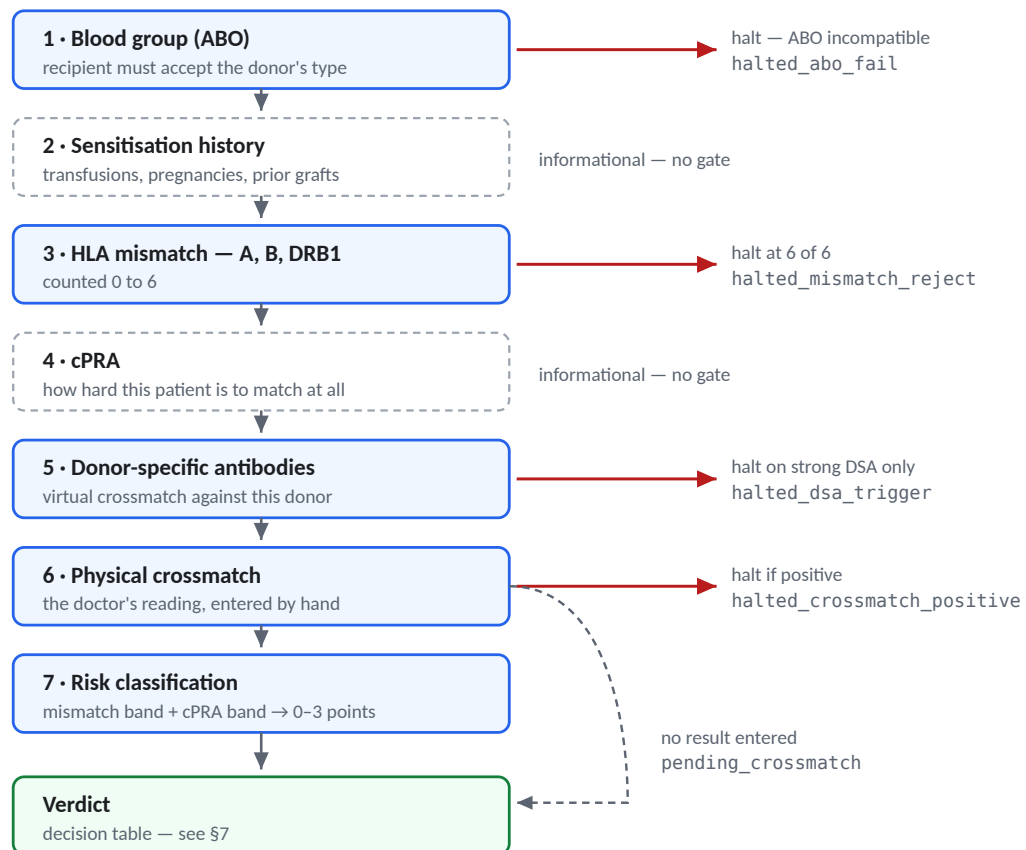


Figure 2 — The pipeline. Solid blue steps are gates; dashed grey steps compute values that inform the report but never stop it. A halt at any gate means every later step is recorded as "not evaluated" rather than as passing.

The Living Kidney Donor Profile Index is computed only when the pipeline reaches step 6 or beyond. Every halted report carries no LKDPI at all — there is no point scoring donor quality for a donation that cannot happen.

4 Steps 1–3

4.1 Blood group

A direct table lookup. The recipient's blood type determines which donor types are acceptable. Rhesus factor is not used — RhD is not expressed on renal vascular endothelium in a way that drives rejection.

RECIPIENT	ACCEPTABLE DONOR TYPES
O	O
A	A, O
B	B, O
AB	A, B, AB, O

4.2 Sensitisation history

Prior transfusions, pregnancies and previous transplants each carry a weight; the pipeline computes a sensitisation score and an adjusted MFI cutoff from them. **Neither currently gates anything.** The value appears in the report labelled "Computed for reference — not yet a reject/proceed gate", which is the honest description of its status.

4.3 HLA mismatch

Counted across three loci — **A, B and DRB1** — as the number of donor alleles the recipient does not also carry:

```
mismatches at locus = | set(donor alleles) - set(patient alleles) |
```

A set difference, so a homozygous donor (for instance 33/33) contributes at most one mismatch at that locus. With two alleles across three loci the total runs from 0 to 6.

TOTAL	BAND	MEANING
0	0 mismatches	Identical at all three counted loci
1–2	<3 mismatches	
3–6	3–6 mismatches	
6	halt	Rejected — see below

Rejection is at **6 or more**, inclusive. Because 6 is also the maximum the count can reach, a strict "more than 6" rule would have been dead code and never fired — a real defect that was found and corrected.

Missing typing is scored at the worst case of 2 mismatches for that locus, and clears a data-completeness flag. That flag is what later produces a *cannot assess* verdict rather than a falsely confident one: the system will not tell you a pairing is compatible on typing it never saw.

5 Steps 4–6

5.1 cPRA — calculated panel reactive antibody

cPRA answers a different question from the rest of the pipeline. It is not about this donor: it estimates the percentage of the *general donor population* this patient would react against. A cPRA of 80% means roughly four in five potential donors are unavailable to them.

Antibodies above **2,000 MFI** are taken as clinically relevant sensitisations. Each one's antigen is looked up in the Sri Lankan population frequency table, and the frequencies are combined as a union of independent events:

```
P_combined ← P_combined + f - (P_combined × f)      for each antigen's frequency f
cPRA %      = P_combined × 100
```

The formula asks "what fraction of donors carry *at least one* of the antigens this patient reacts to?" Each new antigen adds only the part of the population not already excluded — which is why frequencies are combined this way rather than

summed. Two antigens at 20% and 10% give $0.2 + 0.1 - 0.02 = 28\%$, not 30%.

Duplicate antigens are removed before combining, and an antigen absent from the reference table contributes zero rather than a guess. The report states how many of the patient's sensitisations were found in the table, so an unusually low cPRA can be recognised as thin data rather than good news.

CPRA	BAND	EFFECT
< 30%	<30%	0 risk points
30–60%	30-60%	1 risk point
> 60%	>60%	No agreed point value — see §6

5.2 Donor-specific antibodies — the virtual crossmatch

The patient's antibody profile is checked against this donor's HLA antigens. Any antibody at or above 1,000 MFI targeting an antigen the donor carries is a donor-specific antibody, graded into three bands. **Only a strong DSA halts the check.**

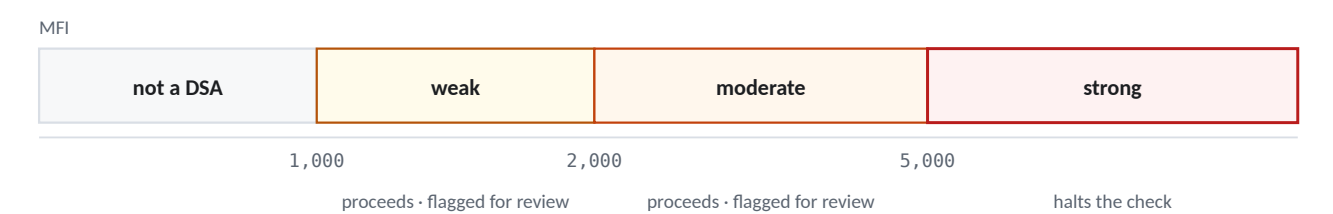


Figure 3 — DSA severity bands. A weak or moderate antibody leaves the pairing viable but produces a *proceed with caution* verdict with a desensitisation-review flag. Only strong antibodies stop the pipeline.

Matching an antibody to a donor antigen requires the two naming conventions to meet. Donor typing is stored molecularly (DRB1*13) while bead charts report serological names (DR13), so donor alleles are converted using the standard prefix mapping — DRB1 to DR, DQB1 to DQ, DPB1 to DP, C to Cw — and antibody names have their Bw suffix stripped, so B45,Bw6 matches B45 .

5.3 Physical crossmatch

Unlike every other step, this value is not computed and not read from stored data. It is the doctor's own reading of a laboratory result, re-entered by hand on the review step of every check. The pair record stores a transcription of the lab report for reference, but the pipeline never reads it — a deliberate choice, recorded in the code:

"Deliberately no crossmatch_is_positive column — that's the doctor's own reading, re-entered fresh on every compatibility check's Review step, never OCR-filled and never stored here."

A positive crossmatch halts. If no result is entered at all, the report ends at *awaiting crossmatch* — a complete record of everything checked so far, explicitly not a verdict.

6 Step 7 — the risk level

This is the only place in the verdict path where numbers are added together. Two bands each contribute points, and the sum maps to a label:

HLA MISMATCH BAND	POINTS	CPRA BAND	POINTS
0 mismatches	0	<30%	0
<3 mismatches	1	30–60%	1
3–6 mismatches	2	>60%	–

SUM	FINAL RISK LEVEL
0	Low Risk
1	Low-Average Risk
2	High-Average Risk
3	High Risk

WHY A CPRA ABOVE 60% PRODUCES NO RISK LEVEL

No point value was agreed for the >60% band, so rather than inventing one the system returns **no risk level at all** and the verdict becomes *proceed with caution* with a high-cPRA flag. A highly sensitised patient is precisely the case where a fabricated number would be most misleading. The same applies when HLA typing is incomplete: no risk level is produced, because two of the three loci cannot honestly be scored.

A consequence worth stating: a meaningful share of real reports will carry no final risk level. That is the system declining to answer, not a gap in the calculation.

7 The decision table

Six rules, evaluated in order. The first that matches determines the verdict.

#	CONDITION	VERDICT	ACTION GIVEN
1	The pipeline halted at any gate	Not compatible	"This donor cannot proceed. Add the patient to the paired exchange pool, or search for another donor."
2	No crossmatch result was entered	Cannot assess	"Submit a crossmatch result and re-run the check."
3	Completed, but HLA typing was incomplete	Cannot assess	Names the exact missing typing, e.g. "Enter the donor DRB1 typing, then re-run the check."
4	Completed, but no risk level could be assigned	Proceed with caution	"Refer to the desensitisation protocol before scheduling."
5	Completed, with one or more review flags	Proceed with caution	As above, carrying the risk level
6	Completed, nothing flagged	Compatible	None. "This pairing cleared every gate with a final risk level of ..."

Every verdict also records **which step decided it** — "determined at step 3 of 7" — so a clinician can see immediately whether a pairing failed at the first hurdle or the last.

Review flags

FLAG	RAISED WHEN
Incomplete typing	A locus was missing on either side and was worst-cased
DSA requires review	A weak or moderate donor-specific antibody was found
High cPRA	The patient's cPRA exceeds 60%
Unclassified risk	No final risk level could be assigned

A NOTE ON ORDERING

Because rule 3 precedes rule 4, an incomplete-typing report is reported as *cannot assess* rather than *proceed with caution* — missing data is never dressed up as a clinical judgement. Conversely rule 4 precedes rule 5, so a highly sensitised patient always reaches *proceed with caution* and never *cannot assess*: the system has the data, it simply declines to grade it. Both orderings are covered by dedicated tests.

8 LKDPI — a separate index

The Living Kidney Donor Profile Index estimates how a given living donor's kidney compares with others, from thirteen donor and donor–recipient characteristics. **Higher is worse**. It is displayed alongside the verdict but **never contributes to it** — a fact covered by its own test.

8.1 The model

TERM	COEFFICIENT	APPLIED
Intercept	-11.30	Always
Donor age above 50	+1.85	Per year over 50; nothing at or below 50
Donor eGFR	-0.381	Per mL/min/1.73m ²
Donor BMI	+1.17	Per unit
Donor African-American	+22.34	Binary
Donor ever smoked	+14.33	Binary — former or current
Donor systolic BP	+0.44	Per mmHg
Both male	-21.68	Binary — donor and recipient both male
ABO incompatible	+27.30	Binary
Biologically unrelated	-10.61	Binary
HLA-B mismatches	+8.57	Per mismatch
HLA-DR mismatches	+8.26	Per mismatch
Donor:recipient weight ratio	-50.87	× the ratio, capped at 0.9

Bands: below 0 excellent, below 20 good, below 40 moderate, otherwise marginal. These boundaries are a presentation convention anchored to the published distribution — **not clinical policy**.

If any required input is missing the score is **withheld entirely**. No default, no cohort mean, no partial estimate — the report names what is missing instead. Values outside the ranges the model was built on are flagged but never clamped.

8.2 A worked calculation

A 60-year-old white male donor, never smoked, eGFR 90, BMI 25.0, systolic BP 130, 70 kg, biologically related, ABO compatible, one HLA-B and one HLA-DR mismatch, donating to a 70 kg female recipient:

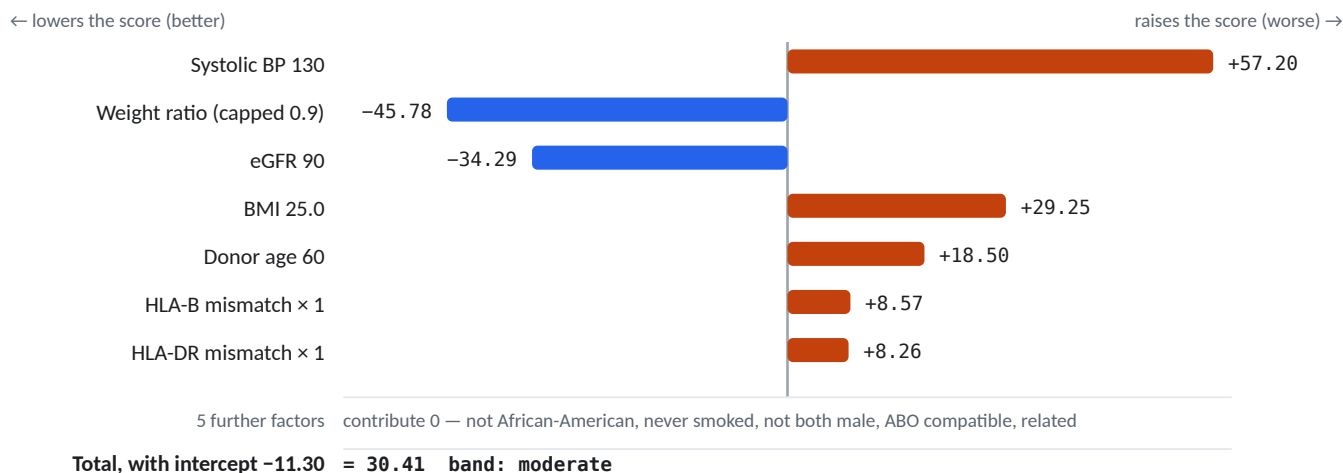


Figure 4 — What moves an LKDPI score. Contributions sorted by magnitude; the intercept is excluded from the bars, so they sum to the score minus the intercept (41.71). Direction carries the meaning as well as colour, so the chart survives being printed in greyscale.

The system also detects when a **single factor dominates**. Each term is compared against a reference donor (age 45, eGFR 95, BMI 26, systolic BP 120, weight ratio 0.9, one B and one DR mismatch, all binary factors absent); if any one term deviates by more than 15 points, the report says so explicitly — "one factor is driving this score". In the example above, a systolic BP of 160 instead of 120 would deviate by 17.6 points and trigger that callout.

USE WITH CAUTION — STATED IN THE MODEL'S OWN DOCUMENTATION

LKDPI's discrimination is modest: a C-statistic of 0.59 in the derivation cohort of 36,025, falling to **0.55** in both a European (n=416) and a Canadian (n=645) validation, with a Japanese cohort (n=133) finding no association with three-year graft survival. **It has never been validated in any South Asian population.** Its only race term is binary African-American versus not, so every locally registered donor scores zero on it — an untested extrapolation rather than a validated category. Its BMI coefficient was calibrated on US body habitus, where South Asian metabolic risk begins at a lower BMI. This is why it informs, and never decides.

9 Worked examples

Four cases, taken from the integration test suite — these run on every commit, so the numbers below are guaranteed to match the system's behaviour.

Case 1 — clears every gate

Recipient AB, donor O. No antibody profile. Crossmatch negative.

STEP	RESULT
1 • ABO	AB accepts O — pass
2 • Sensitisation	No events; score 0.0, adjusted cutoff 2,000
3 • Mismatch	A: 29/33 vs 33/33 → 0 • B: 07/58 vs 40/40 → 1 • DRB1: 03/04 vs 13/14 → 2 • total 3, band "3-6"
4 • cPRA	0.0%, band "<30%"
5 • DSA	None found
6 • Crossmatch	Negative — pass
7 • Risk	2 points (3-6 mismatches) + 0 (<30% cPRA) = 2 → High-Average Risk
Verdict	Compatible • no flags • no action required

Case 2 — halts on mismatch

Recipient AB, donor O; patient A 01/02, B 07/08, DRB1 03/04 against donor A 11/12, B 17/18, DRB1 13/14. Every allele differs: $2 + 2 + 2 = 6$, which meets the rejection threshold.

The check stops at step 3. cPRA and risk level are null, not zero — the distinction matters, and the report shows steps 4 to 7 as "not evaluated". Verdict **Not compatible**, determined at step 3 of 7.

Case 3 — halts on a strong antibody

Recipient O, donor O, identical HLA typing on both sides — zero mismatches, an otherwise excellent pairing. The patient carries an antibody against **B7** at **6,000 MFI**, and the donor's B allele **07** converts to the designation **B7**. At 6,000 the antibody is graded strong, and the check halts at step 5:

"CRITICAL WARNING: Donor-Specific Antibody (DSA) detected. Patient has a strong antibody against donor HLA B7 with an MFI of 6000. Process halted due to very high risk of rejection."

A well-matched pairing on paper, correctly stopped. This is the case that most justifies the virtual crossmatch existing at all.

Case 4 — proceeds, with a flag

The same pairing as case 1, but the patient carries an antibody against B40 at **3,500 MFI** and the donor's B alleles are 40/40. The antibody matches, but 3,500 falls in the moderate band — so the pipeline does not halt. It completes with a review flag, and the verdict is **Proceed with caution** with the action "Refer to the desensitisation protocol before scheduling."

10 What the clinician sees

The report screen presents the verdict first and the evidence beneath it.

1. **Verdict hero** — the verdict in large type, the one-sentence explanation, a risk-level badge where one exists, and "determined at step n of 7".

2. **Action strip** — the single next step, when there is one. Hidden entirely when the pairing is clear, so its presence is itself a signal.
3. **Review flags** — one card per flag, each with a plain-language explanation.
4. **LKDPI card** — shown unless the pairing was ruled out. The score, its band, a plain-English sentence, the single-factor callout if one fired, any out-of-range warnings, and the contribution chart from §8.2 with a "show all factors" toggle. Where inputs were missing, the card explains what is absent and states that the score was withheld rather than estimated.
5. **Step timeline** — all seven steps as an accordion, each with a pass, halt, flag or not-evaluated marker, expandable to the underlying numbers: the per-locus mismatch table, each DSA match with its MFI and severity, the crossmatch T- and B-cell readings, and so on.

The timeline is built from the raw stored step results rather than from the summarised verdict, so what a clinician expands is the actual computation, not a retelling of it.

11 What is stored, and why

Each report keeps every step's full result as structured data, plus the verdict. Reports are write-once: there is no update path, and no function anywhere in the codebase modifies one after creation.

Crucially, the report also **snapshots the reference data that was in force** when it was generated — the DSA floor and band boundaries, the HLA frequency table's version string and sample size, the LKDPI citation and its limitation note, and the entire computed verdict.

WHY SNAPSHOT

From the code: the verdict is "computed once and stored alongside the raw step results rather than re-derived on every read, so a later change to the decision table doesn't silently reinterpret an old report." If the DSA bands are revised next year, a report generated today must continue to mean what it meant today. Without this, revisiting a historical clinical decision would silently show a different answer than the one that was acted upon.

The report and its audit-log entry are written in a single database transaction, so a report can never exist without a record of who generated it. Where a doctor runs a check against another hospital's donor, the audit entry records that explicitly as a cross-hospital check.

12 Reference data and provenance

CONSTANT SET	SOURCE	STANDING
ABO compatibility	OPTN blood group compatibility policy	Standard clinical convention
HLA antigen frequencies	Grifoni et al., <i>Hum Immunol.</i> 2018;79(2):87–88 · PMID 29289740 · AFND population 3423. 714 blood-bank donors, Colombo, all 26 districts. 97 antigens across six loci. Version grifoni-colombo-2018-v1	Population-specific and well sourced — computed by direct counting from the raw genotype export, not transcribed from the paper
LKDPI coefficients	Massie et al., <i>Am J Transplant</i> 2016;16(7):2077–2084 · all thirteen verified against the full text	Published model, weak external validation, none in South Asia — see §8
DSA thresholds	Project-internal; replaced an earlier flat 1,000 MFI threshold	No published citation. Snapshotted per report
Mismatch bands	Project specification, from clinician input	No external citation
cPRA bands and risk points	Project specification, from clinician input	No external citation. No point value agreed above 60%

The distinction in the right-hand column is worth attention. The HLA frequency table and the LKDPI coefficients are externally published and traceable. The band boundaries and point values that turn them into a risk level are **local convention agreed with clinicians**, not literature. They are reasonable, they are documented, and they should be reviewed by the transplant team as policy rather than inherited as fact.

13 Known limitations and defects

13.1 Acknowledged in the code

- **cPRA assumes antigens occur independently.** HLA is inherited in haplotypes, so a patient sensitised to one antigen is more likely to be sensitised to those inherited alongside it. This **inflates** computed cPRA. The correction needs the study's haplotype table, which could not be obtained.
- **Two steps compute values that gate nothing.** Sensitisation history and the cPRA band are informational. The cPRA result even carries a halt field that is never acted on — easy to misread as active.
- **A legacy nine-locus score and a four-tier "risk tier" are still computed** on completed runs and appear nowhere in the verdict. They use a different vocabulary — "High-Moderate Risk" against step 7's "High-Average Risk" — which is a genuine confusion risk for a clinician reading both.
- **The cPRA sufficiency flag is hard-coded true**, so the interface's "not enough population data" message is currently unreachable.

13.2 Found during review, not yet addressed

ISSUE	EFFECT
Worst-cased mismatch counts feed LKDPI	A locus with no typing scores 2 mismatches, and that value flows into the LKDPI HLA-B and HLA-DR terms unflagged. An <i>awaiting crossmatch</i> report can therefore display an LKDPI computed partly on fabricated mismatch counts. The completeness flag blocks the risk level but not the index. Worth fixing
Antibodies against DRB3/4/5, DQA1 and DPA1 cannot match	Those loci have no serological prefix mapping, so donor antigens from them are generated in a form no antibody name can equal. A genuine antibody against them is invisible to the virtual crossmatch. Worth fixing
Two cutoffs use different comparison operators	cPRA counts antibodies strictly above 2,000; DSA grading includes exactly 1,000. An antibody at exactly 2,000.00 is graded a moderate DSA but does not count toward cPRA. Minor
The DSA grading fallback favours severity	An MFI in the microscopic gap between bands would be graded strong. Unreachable through the database, but the fallback errs toward halting rather than toward flagging a data problem. Minor
Antigen-name normalisation is applied at call sites, not at source	The pipeline normalises correctly in both places that matter, but the underlying query returns raw chart strings — so a future caller could silently get it wrong. Refactor
An unrecognised sensitisation event type raises an error	A direct dictionary lookup with no fallback would fail the whole check. Minor

14 Test coverage

The scoring path is the most heavily tested part of the system.

AREA	TESTS	NOTABLE
Full pipeline (integration)	30	All four worked examples in §9; the 6-of-6 threshold regression; antigen-naming bridges for DRB1→DR, C→Cw and the Bw suffix; all five verification refusals, including that a blocked check creates no report at all
LKDPI	15	The 30.41 hand calculation to two decimals; the weight-ratio cap; contributions summing to score minus intercept; the override firing at a deviation of 17.6 and not at the reference case
HLA mismatch	10	6 halts and 5 does not; an untyped donor scores 6 rather than 0; symmetric handling of missing patient and donor data
Decision table	9	Every row by name; the step number for each halt; that a high cPRA must be <i>proceed with caution</i> and never <i>cannot assess</i>
cPRA	9	The union formula ($0.2 + 0.1 - 0.02 = 0.28$); unknown antigens contribute zero; duplicates deduplicated
cPRA banding	9	Boundary values including 29.9995 — a real off-by-one found in review
DSA grading	8	Each band boundary exactly: 1,000 weak · 3,500 moderate · 6,000 strong and halting
Readiness preview	8	Blocking versus informational gaps
Risk classification	5	All four scored combinations; >60% returns nothing rather than failing
ABO	16	The complete 4×4 matrix
Crossmatch	3	Positive halts, negative proceeds
Historical backfill	3	That the migration which added verdicts to existing reports reproduces the live decision table exactly

Roughly 125 tests cover this path, within 831 across the system, all passing. The boundary cases are tested at the boundary — 1,000 exactly, 2,000 exactly, 29.9995, 6 of 6 — which is where scoring code actually goes wrong.